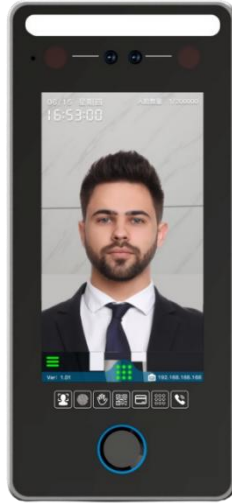


A3-8396 Multimodal facial recognition

access control all-in-one machine



Parameters

Primary applications: Access control, time and attendance, intercom.

Power Supply: DC12V (9~14V)

Recognition Distance: 0.3-2 meters

Recognition Speed: <0.3S

Operating current: <500mA

Size: 192x87x25mm

Screen: 5 inch IPS Full HD screen

Camera: 1MP Dual HD Cameras with Wide Dynamic Range (WDR)

Record capacity: 200,000 records, 1,000 photo records.

Operating environment: Temperature: -30 to 65°C; Humidity: 10 to 90%, non-condensing.

User Capacity: 3000 face、3000 palm-vein、3000 card (IC&NFC card)、3000 pin-code、100 fingerprint

Comparison method: 1:N/1:1

OS: LINUX

Management method: Touch control, network connectivity

Enrollment Methods: Facial recognition, fingerprint, palm vein, password, IC card.

Firmware Upgrade: Supports OTA upgrades and USB flash drive firmware upgrades.

Storage Capacity: 128MB DDR2 RAM, 2GB eMMC storage

Installation: Wall-mounting (Optional bracket for mounting on the turnstile.)

Material: ABS body + glass panel

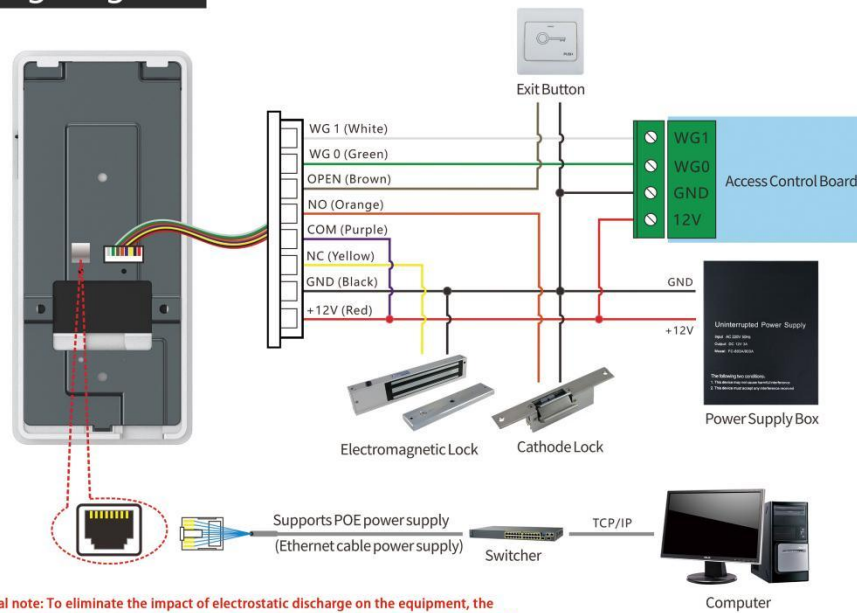
Communication: TCP/IP + WIFI

Algorithms: facial recognition, QR code, fingerprint, palm vein.

Other interfaces: Exit button ×1, door lock control ×1, Wiegand ×1, door contact ×1, alarm input ×1, alarm output ×1.

Wiring Diagram

Wiring Diagram



Special note: To eliminate the impact of electrostatic discharge on the equipment, the GND line should be connected to the ground, and the grounding resistance must be $\leq 4\Omega$.

Wiring Requirements:

- AC220V power wires: Use 3X 1.0mm (or above) wire. To prevent leakage, and the ground wire of the power wire is recommended to be grounded.
- Electric lock wires: 2x1.0 mm (or above) wires shall be used. The maximum wire length shall not exceed 20 meters.
- TCP/IP communication wires: Please use standard network cable. The maximum length of the wire should not exceed 100 meters.
- Door sensor or exit button wires: 2 x 0.5 mm (or above) wires shall be used, and the maximum wire length shall not exceed 100 meters.
- Wiegand signal wires: 4 x 0.5 mm (or above) wires shall be used, and the maximum wire length shall not exceed 80 meters.

REMARK

- ▼ Power on, locked (electric bolt lock or electromagnetic lock) correct connection: the positive electrode of the electric lock connects the power supply +12V. The negative electrode of the electric lock connects to the NC of face access controller. The COM of the face access controller connects to the GND of the power supply.
- ▼ Power on, unlocked (electric control lock or cathode lock) correct connection: the positive electrode of the electric lock connects the power supply +12V. The negative electrode of the electric lock connects to the NO of face access controller. The COM of the face access controller connects to the GND of the power supply.

User Manual

Main Interface

Tap the third icon at the bottom of the main interface (indicated by the arrow in the image below) to switch to the "Administrator Verification" screen. Authenticate using the administrator's registered method (face, palm vein, fingerprint, IC card, or password) to access the device's management menu. (For password authentication, tap the second icon at the bottom to bring up the numeric keypad, enter the administrator password, and then tap "Confirm" to access the management menu.) Note: If no password or administrator has been set, the system will not require authentication.



Main Interface



Main menu

User Management

Click "User Management" in the menu to access the user management section, where you can perform operations such as registering users, modifying user details, backing up data, and restoring data.



User Management interface

Enroll Users

Click "Register User" to begin the new user registration process. Enter the ID number and name, then select the desired registration method(s) based on your needs: face, card, password, palm vein, or fingerprint. Once registration is complete, click "Back" in the top-left corner and select "Yes" to save the user.

注册人员	
编号 (必填) :	3
姓名:	请输入用户名: 0-30个字
注册人脸	
注册卡	
注册密码	
掌静脉注册	0/3
注册指纹	0/3

Enrollment Interface

After selecting the option to register a face, the user must position their face directly facing the device screen, ensuring the entire face is clearly visible within the on-screen frame, as shown in the image below:



Enroll Face Interface

When registering the face, please look directly at the screen. To ensure high-quality parameter capture and accurate matching, maintain a natural expression during the process; avoid wearing accessories such as hats or glasses that might interfere with feature capture, ensure your bangs do not obscure your facial features, and refrain from wearing heavy makeup.

After the facial data has been captured, tap "Confirm" to complete the registration; you may then proceed to select other registration methods available on the device.

Select the card registration option; the device will prompt you to read the card. At this point, swipe the card in the device's card-reading area or manually enter the card number.

The facial recognition terminal will detect the card number (as shown in the image below); tap "Confirm" to complete the card registration.



Enroll Card Interface

Select the option to set a registration password; a password input box will appear. Enter the same password twice (4–11 characters in length) and click "Confirm" to complete the password registration, as shown in the figure below:



Enroll Pincode Interface

To register your palm vein, position your palm facing the camera and wait for the enrollment process to complete. A pop-up notification will appear indicating successful registration; click "Confirm" to register the palm vein of your other hand, or click "Cancel" to finish the process. As shown in the figure below:



Palm-Vein enrollment interface

Select the fingerprint registration button to enter the registration interface. Follow the prompts to register your fingerprint by pressing the same finger three times; the fingerprint is successfully registered once this is done. Tap "Confirm" to complete the process. The fingerprint registration interface is shown below:



Fingerprint Enrollment Interface

Modify User

Click "Modify User" to access the user modification interface, then select a registered user to edit their details. (Users are listed by ID number by default; you can navigate through the list using the "<" and ">" buttons at the bottom, or use the search function in the top-right corner to find a specific registered user by ID or name.)



User modification interface

Locate the user to be modified and click on the user entry to view their details. You can modify information such as the user's name, card number, password, fingerprint, palm vein data, access rights, validity period, and door-opening time slots. Adjusting access rights allows you to designate the user as an administrator. You can also re-enroll the user's face or delete the user entirely.



员工详细信息	
编号	1
姓名	张三
卡	
密码	1111
指纹	1
掌静脉	1
权限	用户
有效期	2037-12-31
开门时段	1



删除

Edit User interface

Data Backup

Click "Data Backup" to back up the registered personnel information to a USB flash drive.

Data Recovery

Click "Data Recovery" to restore the personnel information saved on the USB flash drive to the face recognition terminal.

Attendance Management

Click "Attendance Management" in the menu to access the attendance management section, where you can configure attendance settings.



Attendance Rules Interface

Users can customize the "Company Name" displayed on attendance reports and use a USB flash drive to upload or download shift schedules, attendance sheets, and records.

Full-record warning: Users can specify a remaining record capacity threshold (default: 1,000 records) at which the device triggers an alert as storage nears capacity.

Allowable late arrival and early departure times can be configured, up to a maximum of 120 minutes.

Shift scheduling: Select "Attendance Time Slot Settings" to designate the specific time slot, then configure three sets of attendance times.



Shift Scheduling Settings Interface

Intercom Settings



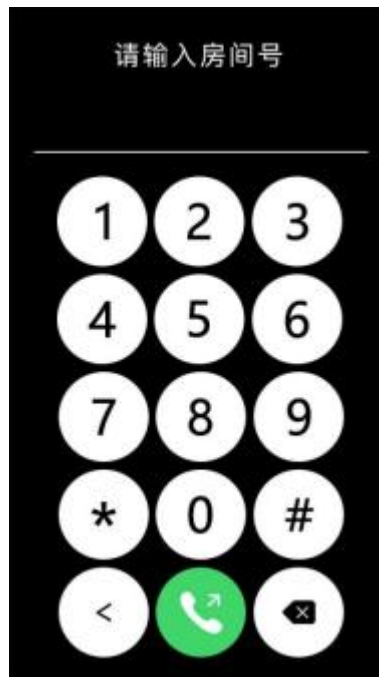
Intercom Settings

The intercom function can be enabled or disabled; once enabled, a phone icon appears in the bottom-right corner of the main interface, allowing for intercom communication with indoor extension units.

You can configure the device name, building number, unit number, service phone number, and security phone number.

You can switch the device's operating mode (e.g., building station or perimeter station).

You can select the intercom dialing method (Direct Call, Dialing, or Building-Specific Direct Call). "Direct Call" broadcasts a call to all extension users associated with the device; "Building-Specific Direct Call" limits calls to residents within the same building; "Dialing" allows you to tap the phone icon on the main interface and enter a specific building and room number to call a particular indoor extension, as shown in the figure below:



Dialer interface

System Settings

Click "System Settings" to enter the system settings menu, which includes Device Settings, Advanced Settings, Recognition Settings, and Alarm Settings.



System Setting Interface

Device Setting



Device Settings Screen

The device's NTP service can be enabled or disabled.

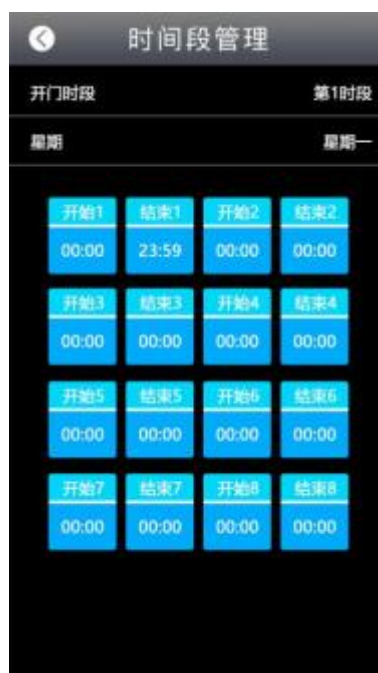
Settings are available for functions such as system time, volume, screen brightness, language, time-slot management, and interface return time.

Volume and screen brightness can be adjusted up to a maximum level of 10.

Language options include Simplified/Traditional Chinese, English, Portuguese, Spanish, Russian, French, Japanese, Korean, Thai, and Arabic.

The interface return time can be set to a maximum of 60 seconds (default is 1 second).

Time-slot management allows for the configuration of door-opening periods from Monday to Sunday, as shown in the figure below.



Time Slot Management Interface

Advanced Settings



Advanced Settings

The total number of device administrators can be modified, with a maximum of 99 administrators supported.

When QR code recognition is enabled, a QR code icon appears in the bottom-left corner of the main interface; tapping it allows for door access via QR code.

All records and user data on the device can be deleted.

The device can be restored to factory settings.

The automatic restart function can be enabled, and the scheduled restart time can be configured.

Device firmware can be upgraded using a USB flash drive.

Recognition Settings



Recognition Settings

The device's liveness detection and recognition thresholds can be adjusted; higher thresholds yield greater recognition accuracy but impose stricter requirements on the quality of registered photos.

The recognition range can be adjusted (near, medium, or far).

Liveness detection, display of the authenticated user's avatar, unauthorized access alerts, and the display of infrared images can be enabled or disabled.

Fill-light mode: The fill light supports three settings—off, always on, and automatic. In automatic mode, the light turns on when a person is detected and turns off when no one is present.

Alarm Settings



Alarm Settings

Enable or disable the device's door sensor alarm and unauthorized verification alarm.

Configure the device's duress password and door-open timeout settings.

Clear alarms: Silence all alarm alerts.

Door Access Control

Click "Access Control Functions" in the menu to enter the access control settings menu, as shown in the figure below:



Door Access Control

Installation location: Configurable as entry or exit point;

Unlock duration is adjustable from 0 to 65,535 (0 represents 0.5 seconds; 65,535 represents "normally open" upon successful verification);

Wiegand output format is configurable to Wiegand 26 (2-byte or 3-byte), Wiegand 34 (4-byte), or disabled;

Supports settings for verification-free entry and multi-person combined verification for access.

View Record

Click "Browse Records" in the menu to enter the record browsing interface. Use the buttons at the bottom to navigate through the records, as shown in the figure below:



照片	登记信息	打卡类型	记录时间
	0001	人脸	2025/09/01 04:14:30
	0002	人脸	2025/09/01 04:14:30
	0003	人脸	2025/09/01 04:14:30
	0004	人脸	2025/09/01 04:14:30
	0005	人脸	2025/09/01 04:14:30
	0006	人脸	2025/09/01 04:14:30
	0007	人脸	2025/09/01 04:14:30
	0008	人脸	2025/09/01 04:14:30

View Record

Network settings

Click "Communication Settings" in the menu to enter the communication settings menu. You can modify device port settings, server settings, and wired network settings, as shown in the figure below:



Communication Settings

Port settings

Click "Port Settings" to enable or disable the device's UDP, TCP, and TCP SSL functions, as well as the display of the IP address on the main interface. You can also configure the UDP port number (default: 8101) and the communication password, as shown in the figure below:



Port Settings

Server Settings

Click "Server Settings" to modify the device's HTTP and all-in-one card services. When the device is installed via the all-in-one card system software and selected for real-time monitoring, the settings here will be automatically populated by the software, as shown in the figure below:



服务器设置界面 Server Setting

Wired Setting

The machine's TCP/IP parameters can be modified. As shown in the figure below:



Wired Network Settings Interface

The default IP address is 192.168.1.150;
the default subnet mask is 255.255.255.000;
the default gateway is 192.168.1.1;
the default DNS server IP is 192.168.1.1;
the MAC address is fixed and unique.

System Info

Click "System Information" in the menu; here, you can view device and registration details for this facial recognition terminal.



System Info

Device Info

Click "Device Information" to view details such as the manufacturer, website, serial number, date of manufacture, device name, firmware version, algorithm version, and font library import status. As shown in the figure below:



Device Info

Registered Info

Click "Registration Info" to view details such as the number of managed items, as well as the number of registered users, faces, fingerprints, palm veins, cards, passwords, total records, and photo records. The number on the left indicates the current count of registered items on the device, while the number on the right indicates the device's maximum capacity for such registrations, as shown in the figure below:



注册信息	
管理登记	0/15
用户登记	4/3000
人脸登记	0/3000
指纹登记	3/100
掌静脉登记	1/3000
卡登记	0/3000
密码登记	2/3000
全部记录	30/200000
拍照记录	29/1000